

# KIT SAFETY DATA SHEET



Kit Product Name Bio-Plex Pro Human Chemokine Panel

Kit Catalogue Number(s) 171AK99MR2, MZ0000XVVLFGWM

Revision date 13-Feb-2024

## Kit Contents

| Catalogue Number(s)                                                 | Product Name                                                   |
|---------------------------------------------------------------------|----------------------------------------------------------------|
| 9723892, 9703892, 9704415, 10014822, 10014823                       | Bio-Plex Assay Buffer                                          |
| 171304040, 10027955, 12006121, 12005850                             | Bio-Plex Pro Assays 10X Wash Buffer                            |
| 171DK0001, 171DK0050, 10032534                                      | Bio-Plex Pro Human Chemokine Standards                         |
| 10032535                                                            | Bio-Plex Pro Human Chemokine Panel 40-plex Lyophilized Control |
| 10032533                                                            | Bio-Plex Pro Human Chemokine Detection Antibodies              |
| 10032444                                                            | Bio-Plex Pro Human Chemokine 40-Plex Beads                     |
| 10032400, 10031831, 12005852                                        | Bio-Plex Detection Antibody Diluent HB                         |
| 10014641, 9704423, 9703895, 171305043, 10022628, 10041561, 12005851 | Bio-Plex Sample Diluent                                        |
| 9703888, 9704424, 171305042, 171304080M, 10022368, 12005853         | Bio-Plex Standard Diluent                                      |
| 171304501, 9704418, 9703887, 9703897                                | Streptavidin-PE                                                |



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 18-Oct-2022

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Assay Buffer  
**Catalogue Number(s)** 9723892, 9703892, 9704415, 10014822, 10014823

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

### Other hazards which do not result in classification

## SECTION 3: Composition/information on ingredients

### Substance

Not applicable

### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Trade secret              | -           | 1 - 2.5      |
| Trade secret              | -           | 0.3 - 0.99   |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.01 - 0.099 |
| Chemical name             | CAS No      | Weight-%     |
| Non-hazardous ingredients | Proprietary | Balance      |

## SECTION 4: First aid measures

### Description of first aid measures

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

### Most important symptoms and effects, both acute and delayed

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

### Indication of any immediate medical attention and special treatment needed

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

## SECTION 5: Firefighting measures

### Suitable Extinguishing Media

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

### Specific hazards arising from the chemical

|                                                   |             |
|---------------------------------------------------|-------------|
| <b>Specific hazards arising from the chemical</b> | None known. |
|---------------------------------------------------|-------------|

### Special protective actions for fire-fighters

|                                         |                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Special protective equipment for</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|-----------------------------------------|-------------------------------------------------------------------------------------------------|

fire-fighters Use personal protection equipment.

## SECTION 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

Personal precautions See section 8 for more information.

For emergency responders Use personal protection recommended in Section 8.

### Environmental precautions

Environmental precautions See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

Prevention of secondary hazards Clean contaminated objects and areas thoroughly observing environmental regulations.

## SECTION 7: Handling and storage

### Precautions for safe handling

Advice on safe handling Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

Storage Conditions Store according to product and label instructions.

Incompatible materials None known based on information supplied.

## SECTION 8: Exposure controls/personal protection

### Control parameters

Exposure Limits This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

Engineering controls Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection             | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| Skin and body protection        | Wear suitable protective clothing.                                                                                                                                       |
| Hand protection                 | Wear suitable gloves.                                                                                                                                                    |
| Respiratory protection          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| Environmental exposure controls | No information available.                                                                                                                                                |

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | aqueous solution         |
| Colour          | colourless               |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | 0 °C              |                         |
| Initial boiling point and boiling range | 100 °C            |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapour pressure                         | No data available | None known              |
| Relative vapour density                 | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| Explosive properties                    | Not applicable    |                         |
| Oxidising properties                    | Not applicable    |                         |
| <b>Other information</b>                |                   |                         |
| Molecular weight                        | Not applicable    |                         |
| VOC content                             | Not applicable    |                         |

**SECTION 10: Stability and reactivity****Reactivity**

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

**Chemical stability**

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

#### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

#### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

#### Incompatible materials

**Incompatible materials** None known based on information supplied.

#### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

### **SECTION 11: Toxicological information**

#### Acute toxicity

#### Information on likely routes of exposure

##### **Product Information**

.

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

#### Numerical measures of toxicity - Product Information

No information available

| Chemical name | Oral LD50            | Dermal LD50              | Inhalation LC50               |
|---------------|----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )   | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )     | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 8290 mg/kg ( Rat ) | > 7940 mg/kg ( Rabbit )  | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret  | = 27 mg/kg ( Rat )   | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |

See section 16 for terms and abbreviations

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Carcinogenicity</b>          | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>    | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>   | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b> | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>        | Based on available data, the classification criteria are not met. |

## SECTION 12: Ecological information

### Ecotoxicity

#### Ecotoxicity

**Unknown aquatic toxicity** 0.095 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                          | -                                                                                                    |

### Persistence and degradability

**Persistence and degradability** No information available.

### Bioaccumulative potential

**Bioaccumulation** No information available.

### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

### Other adverse effects

Other adverse effects No information available.

## SECTION 13: Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## SECTION 14: Transport information

**ADG** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

## SECTION 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### Australia

See section 8 for national exposure control parameters

##### **Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

No poisons schedule number allocated

#### International Inventories

Contact supplier for inventory compliance status

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

## SECTION 16: Other information

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 18-Oct-2022

**Revision Note** Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**



---

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 07-Feb-2024

Revision Number 2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Pro Assays 10X Wash Buffer  
**Catalogue Number(s)** 171304040, 10027955, 12006121, 12005850

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

### Other hazards which do not result in classification

## SECTION 3: Composition/information on ingredients

### Substance

Not applicable

**Mixture**

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Trade secret              | -           | 5 - 10   |
| Non-hazardous ingredients | Proprietary | Balance  |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

|                                                   |             |
|---------------------------------------------------|-------------|
| <b>Specific hazards arising from the chemical</b> | None known. |
|---------------------------------------------------|-------------|

**Special protective actions for fire-fighters**

|                                                       |                                                                                                                                       |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

#### **Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

#### **Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **SECTION 7: Handling and storage**

#### **Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

#### **Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** Metals.

### **SECTION 8: Exposure controls/personal protection**

#### **Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

#### **Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### **Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

Environmental exposure controls No information available.

## SECTION 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | aqueous solution         |
| Colour          | colourless               |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | 100 °C            |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapour pressure                         | No data available | None known              |
| Relative vapour density                 | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| Explosive properties                    | Not applicable    |                         |
| Oxidising properties                    | Not applicable    |                         |
| <u>Other information</u>                |                   |                         |
| Molecular weight                        | Not applicable    |                         |
| VOC content                             | Not applicable    |                         |

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** Metals.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****Acute toxicity****Information on likely routes of exposure****Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

**Numerical measures of toxicity - Product Information**

No information available

**The following values are calculated based on chapter 3.1 of the GHS document**

**ATEmix (oral)** 34,207.50 mg/kg

**Component Information**

| Chemical name | Oral LD50        | Dermal LD50              | Inhalation LC50       |
|---------------|------------------|--------------------------|-----------------------|
| Trade secret  | = 3 g/kg ( Rat ) | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h |

See section 16 for terms and abbreviations

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Based on available data, the classification criteria are not met.

**SECTION 12: Ecological information****Ecotoxicity****Ecotoxicity**

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, Lepomis macrochirus)<br>LC50: =12946mg/L (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L (96h, Pimephales promelas)<br>LC50: =7050mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L (96h, Pimephales promelas)<br>LC50: 4747 - 7824mg/L (96h, Oncorhynchus mykiss) | -                          | EC50: =1000mg/L (48h, Daphnia magna)<br>EC50: 340.7 - 469.2mg/L (48h, Daphnia magna) |

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** No information available.

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

**SECTION 13: Disposal considerations****Disposal methods**

**Waste from residues/unused products** Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information**

ADG Not regulated

IATA Not regulated

IMDG Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

## SECTION 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### Australia

See section 8 for national exposure control parameters

##### **Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

No poisons schedule number allocated

#### International Inventories

Contact supplier for inventory compliance status

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

## SECTION 16: Other information

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 07-Feb-2024

**Revision Note** SDS sections updated. 2. 12.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database



International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 22-Aug-2023

Revision Number 1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine Standards

**Catalogue Number(s)** 171DK0001, 171DK0050, 10032534

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

#### Other hazards which do not result in classification

Contains animal source material (Cattle)

## SECTION 3: Composition/information on ingredients

**Substance**

Not applicable

**Mixture**

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Trade secret              | -           | 5 - 10       |
| Trade secret              | -           | 5 - 10       |
| Acetic acid               | 64-19-7     | 0.01 - 0.099 |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

|                                                   |             |
|---------------------------------------------------|-------------|
| <b>Specific hazards arising from the chemical</b> | None known. |
|---------------------------------------------------|-------------|

**Special protective actions for fire-fighters**

|                                                       |                                                                                                                                       |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## SECTION 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## SECTION 8: Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name          | Australia                                                                              | ACGIH TLV                   |
|------------------------|----------------------------------------------------------------------------------------|-----------------------------|
| Acetic acid<br>64-19-7 | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> | STEL: 15 ppm<br>TWA: 10 ppm |

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection             | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| Skin and body protection        | Wear suitable protective clothing.                                                                                                                                       |
| Hand protection                 | Wear suitable gloves.                                                                                                                                                    |
| Respiratory protection          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| Environmental exposure controls | No information available.                                                                                                                                                |

## SECTION 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                 |                             |
|-----------------|-----------------------------|
| Physical state  | Solid                       |
| Appearance      | powder or cake, lyophilised |
| Colour          | white                       |
| Odour           | Odourless.                  |
| Odour threshold | No information available    |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      | 7.4               |                  |
| Melting point / freezing point          | No data available | None known       |
| Initial boiling point and boiling range | No data available | None known       |
| Flash point                             | No data available | None known       |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapour pressure                         | No data available | None known       |
| Relative vapour density                 | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Soluble in water  |                  |
| Solubility(ies)                         | No data available | None known       |
| Partition coefficient                   | No data available | None known       |
| Autoignition temperature                | No data available | None known       |
| Decomposition temperature               |                   | None known       |
| Kinematic viscosity                     | No data available | None known       |
| Dynamic viscosity                       | No data available | None known       |
| Explosive properties                    | Not applicable    |                  |
| Oxidising properties                    | Not applicable    |                  |

### Other information

|                  |                |
|------------------|----------------|
| Molecular weight | Not applicable |
| VOC content      | Not applicable |

## SECTION 10: Stability and reactivity

### Reactivity

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

### Chemical stability

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

### Explosion data

Sensitivity to mechanical impact None.

**Sensitivity to static discharge** None.

#### **Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

#### **Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

#### **Incompatible materials**

**Incompatible materials** None known based on information supplied.

#### **Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

### **SECTION 11: Toxicological information**

#### **Acute toxicity**

#### **Information on likely routes of exposure**

##### **Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

#### **Numerical measures of toxicity - Product Information**

The following values are calculated based on chapter 3.1 of the GHS document

**ATEmix (oral)** 33,333.30 mg/kg

##### **Component Information**

| Chemical name | Oral LD50            | Dermal LD50              | Inhalation LC50         |
|---------------|----------------------|--------------------------|-------------------------|
| Trade secret  | = 3 g/kg ( Rat )     | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h   |
| Trade secret  | = 17 g/kg ( Rat )    | -                        | -                       |
| Acetic acid   | = 3310 mg/kg ( Rat ) | = 1060 mg/kg ( Rabbit )  | = 11.4 mg/L ( Rat ) 4 h |

See section 16 for terms and abbreviations

#### **Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Carcinogenicity</b>          | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>    | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>   | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b> | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>        | Based on available data, the classification criteria are not met. |

## SECTION 12: Ecological information

### Ecotoxicity

#### Ecotoxicity

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Acetic acid   | -                    | LC50: =79mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =75mg/L (96h, <i>Lepomis macrochirus</i> )                                                                                                                                                                                                                                                      | -                          | EC50: =65mg/L (48h, <i>Daphnia magna</i> )                                                           |

### Persistence and degradability

**Persistence and degradability** No information available.

### Bioaccumulative potential

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Acetic acid   | -0.17                 |

### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

### Other adverse effects

Other adverse effects No information available.

## SECTION 13: Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## SECTION 14: Transport information

**ADG** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

## SECTION 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### Australia

See section 8 for national exposure control parameters

##### **Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)

**Poison Schedule Number** 6

##### **National pollutant inventory**

Subject to reporting requirement

| Chemical name         | National pollutant inventory     |
|-----------------------|----------------------------------|
| Acetic acid - 64-19-7 | 10 tonne/yr Threshold category 1 |

##### International Inventories

Contact supplier for inventory compliance status

##### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

## SECTION 16: Other information

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety



Revision date 22-Aug-2023

Revision Note Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 22-Aug-2023

Revision Number 1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine Panel 40-plex Lyophilized Control

**Catalogue Number(s)** 10032535

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

#### Other hazards which do not result in classification

Contains animal source material (Cattle)

## SECTION 3: Composition/information on ingredients

**Substance**

Not applicable

**Mixture**

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Trade secret              | -           | 5 - 10       |
| Trade secret              | -           | 5 - 10       |
| Acetic acid               | 64-19-7     | 0.01 - 0.099 |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures**

**Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures**

**Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

|                                                   |             |
|---------------------------------------------------|-------------|
| <b>Specific hazards arising from the chemical</b> | None known. |
|---------------------------------------------------|-------------|

**Special protective actions for fire-fighters**

|                                                       |                                                                                                                                       |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## SECTION 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## SECTION 8: Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name          | Australia                                                                              | ACGIH TLV                   |
|------------------------|----------------------------------------------------------------------------------------|-----------------------------|
| Acetic acid<br>64-19-7 | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> | STEL: 15 ppm<br>TWA: 10 ppm |

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection             | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| Skin and body protection        | Wear suitable protective clothing.                                                                                                                                       |
| Hand protection                 | Wear suitable gloves.                                                                                                                                                    |
| Respiratory protection          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| Environmental exposure controls | No information available.                                                                                                                                                |

## SECTION 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                 |                             |
|-----------------|-----------------------------|
| Physical state  | Solid                       |
| Appearance      | powder or cake, lyophilised |
| Colour          | white                       |
| Odour           | Odourless.                  |
| Odour threshold | No information available    |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      | 7.4               |                  |
| Melting point / freezing point          | No data available | None known       |
| Initial boiling point and boiling range | No data available | None known       |
| Flash point                             | No data available | None known       |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapour pressure                         | No data available | None known       |
| Relative vapour density                 | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Soluble in water  |                  |
| Solubility(ies)                         | No data available | None known       |
| Partition coefficient                   | No data available | None known       |
| Autoignition temperature                | No data available | None known       |
| Decomposition temperature               |                   | None known       |
| Kinematic viscosity                     | No data available | None known       |
| Dynamic viscosity                       | No data available | None known       |
| Explosive properties                    | Not applicable    |                  |
| Oxidising properties                    | Not applicable    |                  |

### Other information

|                  |                |
|------------------|----------------|
| Molecular weight | Not applicable |
| VOC content      | Not applicable |

## SECTION 10: Stability and reactivity

### Reactivity

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

### Chemical stability

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information**

**Acute toxicity**

**Information on likely routes of exposure**

**Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

**Numerical measures of toxicity - Product Information**

The following values are calculated based on chapter 3.1 of the GHS document

**ATEmix (oral)** 33,333.30 mg/kg

**Component Information**

| Chemical name | Oral LD50            | Dermal LD50              | Inhalation LC50         |
|---------------|----------------------|--------------------------|-------------------------|
| Trade secret  | = 3 g/kg ( Rat )     | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h   |
| Trade secret  | = 17 g/kg ( Rat )    | -                        | -                       |
| Acetic acid   | = 3310 mg/kg ( Rat ) | = 1060 mg/kg ( Rabbit )  | = 11.4 mg/L ( Rat ) 4 h |

See section 16 for terms and abbreviations

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| Germ cell mutagenicity   | Based on available data, the classification criteria are not met. |
| Carcinogenicity          | Based on available data, the classification criteria are not met. |
| Reproductive toxicity    | Based on available data, the classification criteria are not met. |
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## SECTION 12: Ecological information

### Ecotoxicity

#### Ecotoxicity

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Acetic acid   | -                    | LC50: =79mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =75mg/L (96h, <i>Lepomis macrochirus</i> )                                                                                                                                                                                                                                                      | -                          | EC50: =65mg/L (48h, <i>Daphnia magna</i> )                                                           |

### Persistence and degradability

**Persistence and degradability** No information available.

### Bioaccumulative potential

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Acetic acid   | -0.17                 |

### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

**SECTION 13: Disposal considerations**

**Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information**

**ADG** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

**SECTION 15: Regulatory information**

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National regulations**

**Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)

**Poison Schedule Number** 6

**National pollutant inventory**

Subject to reporting requirement

| Chemical name         | National pollutant inventory     |
|-----------------------|----------------------------------|
| Acetic acid - 64-19-7 | 10 tonne/yr Threshold category 1 |

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable



## SECTION 16: Other information

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 22-Aug-2023

**Revision Note** Reformatted and updated existing information.

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 22-Aug-2023

Revision Number 1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine Detection Antibodies

**Catalogue Number(s)** 10032533

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

|                           |                      |
|---------------------------|----------------------|
| <b>Skin sensitisation</b> | Category 1A - (H317) |
|---------------------------|----------------------|

### Label elements

Exclamation mark



**Signal word**

Warning

**Hazard statements**

H317 - May cause an allergic skin reaction

**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapours/spray

Contaminated work clothing should not be allowed out of the workplace

Wear protective gloves/protective clothing/eye protection/face protection

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention

Take off all contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

Harmful to aquatic life with long lasting effects Harmful to aquatic life Contains animal source material (Cattle)

**General Hazards**

Contains human source material and / or potentially infectious components

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Animal Source Antibody    | NO-CAS-90   | 5 - 10       |
| Trade secret              | -           | 0.3 - 0.99   |
| Sodium phosphate dibasic  | 7558-79-4   | 0.1 - 0.299  |
| Human Source Material     | NO-CAS-20   | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.001 - 0.01 |
| Trade secret              | -           | 0.001 - 0.01 |
| Antibodies                | NO-CAS-81   | 0.001 - 0.01 |
| Trade secret              | -           | 0.001 - 0.01 |
| Trade secret              | -           | 0.001 - 0.01 |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Emergency telephone number**

Poisons Information Centre, Australia: 13 11 26

Poisons Information Centre, New Zealand: 0800 764 766

**Inhalation**

Remove to fresh air.

**Eye contact**

Contains human source material and / or potentially infectious components. Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor.

**Skin contact**

Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor. Wash skin with soap and water.

**Ingestion** Call a doctor. Contains human source material and / or potentially infectious components.

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors** May cause sensitisation in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

## **SECTION 5: Firefighting measures**

**Suitable Extinguishing Media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** Product is or contains a sensitiser. May cause sensitisation by skin contact.

**Special protective actions for fire-fighters**

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## **SECTION 6: Accidental release measures**

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Clean contaminated surface thoroughly. Use: Disinfectant.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## **SECTION 7: Handling and storage**

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations** Follow universal and standard precautions for handling potentially infectious materials.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## **SECTION 8: Exposure controls/personal protection**

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## **SECTION 9: Physical and chemical properties**

**Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** aqueous solution  
**Colour** colourless  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u> | <u>Values</u> | <u>Remarks • Method</u> |
|-----------------|---------------|-------------------------|
| pH              | 7.4           |                         |

|                                         |                   |            |
|-----------------------------------------|-------------------|------------|
| Melting point / freezing point          | No data available | None known |
| Initial boiling point and boiling range | > 100 °C          |            |
| Flash point                             | No data available | None known |
| Evaporation rate                        | No data available | None known |
| Flammability                            | No data available | None known |
| Flammability Limit in Air               |                   | None known |
| Upper flammability or explosive limits  | No data available |            |
| Lower flammability or explosive limits  | No data available |            |
| Vapour pressure                         | No data available | None known |
| Relative vapour density                 | No data available | None known |
| Relative density                        | No data available | None known |
| Water solubility                        | Miscible in water |            |
| Solubility(ies)                         | No data available | None known |
| Partition coefficient                   | No data available | None known |
| Autoignition temperature                | No data available | None known |
| Decomposition temperature               |                   | None known |
| Kinematic viscosity                     | No data available | None known |
| Dynamic viscosity                       | No data available | None known |
| Explosive properties                    | Not applicable    |            |
| Oxidising properties                    | Not applicable    |            |
| <b>Other information</b>                |                   |            |
| Molecular weight                        | Not applicable    |            |
| VOC content                             | Not applicable    |            |

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## SECTION 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

**Product Information**

|                     |                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Skin contact</b> | May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available                                                                                                                                                           |
| <b>Symptoms</b>     | Itching. Rashes. Hives.                                                                                                                                                                                                    |

**Numerical measures of toxicity - Product Information****Component Information**

| Chemical name            | Oral LD50            | Dermal LD50              | Inhalation LC50               |
|--------------------------|----------------------|--------------------------|-------------------------------|
| Water                    | > 90 mL/kg ( Rat )   | -                        | -                             |
| Trade secret             | = 3 g/kg ( Rat )     | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Sodium phosphate dibasic | = 17 g/kg ( Rat )    | -                        | -                             |
| Trade secret             | = 8290 mg/kg ( Rat ) | > 7940 mg/kg ( Rabbit )  | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret             | = 27 mg/kg ( Rat )   | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| Trade secret             | = 2600 mg/kg ( Rat ) | -                        | -                             |
| Trade secret             | = 3200 mg/kg ( Rat ) | -                        | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret             | = 53 mg/kg ( Rat )   | = 87.12 mg/kg ( Rabbit ) | -                             |

See section 16 for terms and abbreviations

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitisation</b> | May cause sensitisation by skin contact.                          |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

**SECTION 12: Ecological information**

**Ecotoxicity**

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Unknown aquatic toxicity** 13.12099 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants                                   | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                                                      | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                                                      | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                          | -                                                                                                    |
| Trade secret  | EC50: =2500mg/L (72h, <i>Desmodesmus subspicatus</i> ) | LC50: =1060mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 750 - 1020mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                                                                             | -                          | EC50: =825mg/L (48h, <i>Daphnia magna</i> )<br>EC50: =83mg/L (48h, <i>Daphnia magna</i> )            |

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** There is no data for this product.

**Component Information**

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | 0.7                   |

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

**SECTION 13: Disposal considerations****Disposal methods**



**Waste from residues/unused products**

Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging**

Do not reuse empty containers.

**SECTION 14: Transport information****ADG**

Not regulated

**IATA**

Not regulated

**IMDG**

Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)

**Poison Schedule Number**

6

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable**The Stockholm Convention on Persistent Organic Pollutants** Not applicable**The Rotterdam Convention** Not applicable**SECTION 16: Other information****Prepared By**

Bio-Rad Laboratories, Environmental Health and Safety

**Revision date**

22-Aug-2023

**Revision Note**

Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend** Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 22-Aug-2023

Revision Number 1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine 40-Plex Beads

**Catalogue Number(s)** 10032444

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

|                           |                      |
|---------------------------|----------------------|
| <b>Skin sensitisation</b> | Category 1A - (H317) |
|---------------------------|----------------------|

### Label elements

Exclamation mark



**Signal word**

Warning

**Hazard statements**

H317 - May cause an allergic skin reaction

**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapours/spray

Contaminated work clothing should not be allowed out of the workplace

Wear protective gloves/protective clothing/eye protection/face protection

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention

Take off all contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

Harmful to aquatic life with long lasting effects Harmful to aquatic life Contains animal source material (Cattle)

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Magnetic Beads            | NO-CAS-23   | 20 - 35      |
| Trade secret              | -           | 10 - 20      |
| Trade secret              | -           | 5 - 10       |
| Trade secret              | -           | 5 - 10       |
| Trade secret              | -           | 1 - 2.5      |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Antibodies                | NO-CAS-81   | 0.1 - 0.299  |
| Trade secret              | -           | 0.001 - 0.01 |
| Trade secret              | -           | 0.001 - 0.01 |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Emergency telephone number**

Poisons Information Centre, Australia: 13 11 26

Poisons Information Centre, New Zealand: 0800 764 766

**Inhalation**

Remove to fresh air.

**Eye contact**

Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor.

**Skin contact**

Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor.

**Ingestion**

Rinse mouth thoroughly with water.

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors** May cause sensitisation in susceptible persons. Treat symptomatically.

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** Product is or contains a sensitiser. May cause sensitisation by skin contact.

**Special protective actions for fire-fighters**

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**SECTION 7: Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

**SECTION 8: Exposure controls/personal protection****Control parameters****Exposure Limits**

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | TWA: 10 mg/m <sup>3</sup>                     | TWA: 10 mg/m <sup>3</sup>                                                              |
| Trade secret  | Peak: 2 mg/m <sup>3</sup>                     | Ceiling: 2 mg/m <sup>3</sup>                                                           |
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** Dilute bead suspension in aqueous solution  
**Colour** Cloudy  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C          |                         |
| Flash point                             | No data available | None known              |

|                                        |                    |            |
|----------------------------------------|--------------------|------------|
| Evaporation rate                       | No data available  | None known |
| Flammability                           | No data available  | None known |
| Flammability Limit in Air              |                    | None known |
| Upper flammability or explosive limits | No data available  |            |
| Lower flammability or explosive limits | No data available  |            |
| Vapour pressure                        | No data available  | None known |
| Relative vapour density                | No data available  | None known |
| Relative density                       | No data available  | None known |
| Water solubility                       | Partially miscible |            |
| Solubility(ies)                        | No data available  | None known |
| Partition coefficient                  | No data available  | None known |
| Autoignition temperature               | No data available  | None known |
| Decomposition temperature              |                    | None known |
| Kinematic viscosity                    | No data available  | None known |
| Dynamic viscosity                      | No data available  | None known |
| Explosive properties                   | Not applicable     |            |
| Oxidising properties                   | Not applicable     |            |
| <b>Other information</b>               |                    |            |
| Molecular weight                       | Not applicable     |            |
| VOC content                            | Not applicable     |            |

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## SECTION 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

### Product Information

|                     |                                                                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Skin contact</b> | May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available                                                                                                                                                            |
| <b>Symptoms</b>     | Itching. Rashes. Hives.                                                                                                                                                                                                     |

#### **Numerical measures of toxicity - Product Information**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 50,476.20 mg/kg

#### **Component Information**

| Chemical name | Oral LD50             | Dermal LD50              | Inhalation LC50               |
|---------------|-----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )    | -                        | -                             |
| Trade secret  | = 29700 mg/kg ( Rat ) | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )      | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 15900 mg/kg ( Rat ) | -                        | -                             |
| Trade secret  | = 325 mg/kg ( Rat )   | = 1350 mg/kg ( Rabbit )  | -                             |
| Trade secret  | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| Trade secret  | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                             |

See section 16 for terms and abbreviations

#### **Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitisation</b> | May cause sensitisation by skin contact.                          |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

## **SECTION 12: Ecological information**

### **Ecotoxicity**



**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                    | LC50: =45.4mg/L (96h, <i>Oncorhynchus mykiss</i> )                                                                                                                                                                                                                                                                                                        | -                          | -                                                                                                    |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                          | -                                                                                                    |

#### Persistence and degradability

**Persistence and degradability** No information available.

#### Bioaccumulative potential

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | 0.7                   |

#### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

#### Other adverse effects

**Other adverse effects** No information available.

### **SECTION 13: Disposal considerations**

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information**

**ADG** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**SECTION 15: Regulatory information**

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National regulations****Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)

**Poison Schedule Number** 6

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 22-Aug-2023

**Revision Note** Significant changes throughout SDS. Review all sections.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend** Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)

Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 17-May-2023

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Detection Antibody Diluent HB

**Catalogue Number(s)** 10032400, 10031831, 12005852

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

#### Other hazards which do not result in classification

Contains animal source material (Cattle)

## SECTION 3: Composition/information on ingredients

**Substance**

Not applicable

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Trade secret              | -           | 5 - 10       |
| Antibodies                | NO-CAS-81   | 1 - 2.5      |
| Trade secret              | -           | 0.3 - 0.99   |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.001 - 0.01 |
| Trade secret              | -           | < 0.001      |
| Trade secret              | -           | < 0.001      |
| Chemical name             | CAS No      | Weight-%     |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** None known.

**Special protective actions for fire-fighters**

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**SECTION 7: Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

**SECTION 8: Exposure controls/personal protection****Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Skin and body protection**                Wear suitable protective clothing.

**Hand protection**                          Wear suitable gloves.

**Respiratory protection**                  No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls**        No information available.

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state**                                Liquid  
**Appearance**                                aqueous solution  
**Colour**                                        colourless  
**Odour**                                         Odourless.  
**Odour threshold**                          No information available

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      | 7.4                  |                                |
| <b>Melting point / freezing point</b>          | No data available    | None known                     |
| <b>Initial boiling point and boiling range</b> | 100 °C               |                                |
| <b>Flash point</b>                             | No data available    | None known                     |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapour pressure</b>                         | No data available    | None known                     |
| <b>Relative vapour density</b>                 | No data available    | None known                     |
| <b>Relative density</b>                        | No data available    | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility(ies)</b>                         | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | No data available    | None known                     |
| <b>Decomposition temperature</b>               |                      | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |
| <b>Explosive properties</b>                    | Not applicable       |                                |
| <b>Oxidising properties</b>                    | Not applicable       |                                |
| <br><b><u>Other information</u></b>            |                      |                                |
| <b>Molecular weight</b>                        | Not applicable       |                                |
| <b>VOC content</b>                             | Not applicable       |                                |

**SECTION 10: Stability and reactivity****Reactivity**

**Reactivity** No information available.

#### Chemical stability

**Stability** Stable under normal conditions.

#### **Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

#### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

#### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

#### Incompatible materials

**Incompatible materials** None known based on information supplied.

#### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

### **SECTION 11: Toxicological information**

#### Acute toxicity

#### Information on likely routes of exposure

##### **Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

#### Numerical measures of toxicity - Product Information

| Chemical name | Oral LD50             | Dermal LD50              | Inhalation LC50               |
|---------------|-----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )    | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )      | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 8290 mg/kg ( Rat )  | > 7940 mg/kg ( Rabbit )  | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret  | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| Trade secret  | = 37000 mg/kg ( Rat ) | -                        | > 5.1 mg/L ( Rat ) 4 h        |
| Trade secret  | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                             |



|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

See section 16 for terms and abbreviations

#### **Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

## **SECTION 12: Ecological information**

### **Ecotoxicity**

#### **Ecotoxicity**

**Unknown aquatic toxicity** 0.095 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                          | -                                                                                                    |

### **Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential****Bioaccumulation** No information available.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | 0.7                   |

**Mobility****Mobility in soil** No information available.**Mobility** No information available.**Other adverse effects****Other adverse effects** No information available.**SECTION 13: Disposal considerations****Disposal methods****Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.**Contaminated packaging** Do not reuse empty containers.**SECTION 14: Transport information****ADG** Not regulated**IATA** Not regulated**IMDG** Not regulated**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

No poisons schedule number allocated

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

## SECTION 16: Other information

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 17-May-2023

**Revision Note** Reformatted and updated existing information.

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 17-May-2023

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Sample Diluent  
**Catalogue Number(s)** 10014641, 9704423, 9703895, 171305043, 10022628, 10041561, 12005851

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

#### Other hazards which do not result in classification

Contains animal source material (Cattle)

## SECTION 3: Composition/information on ingredients

**Substance**

Not applicable

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Trade secret              | -           | 0.3 - 0.99   |
| HBR 1 Blocker             | NO-CAS-18   | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.01 - 0.099 |
| Chemical name             | CAS No      | Weight-%     |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** None known.

**Special protective actions for fire-fighters**

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.  
Use personal protection equipment.

## **SECTION 6: Accidental release measures**

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## **SECTION 7: Handling and storage**

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## **SECTION 8: Exposure controls/personal protection**

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                        |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>             | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Skin and body protection</b>        | Wear suitable protective clothing.                                                                                                                                       |
| <b>Hand protection</b>                 | Wear suitable gloves.                                                                                                                                                    |
| <b>Respiratory protection</b>          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                |

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      | 7.4                  |                                |
| <b>Melting point / freezing point</b>          | 0 °C                 |                                |
| <b>Initial boiling point and boiling range</b> | 100 °C               |                                |
| <b>Flash point</b>                             | No data available    | None known                     |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapour pressure</b>                         | No data available    | None known                     |
| <b>Relative vapour density</b>                 | No data available    | None known                     |
| <b>Relative density</b>                        | No data available    | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility(ies)</b>                         | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | No data available    | None known                     |
| <b>Decomposition temperature</b>               |                      | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |
| <b>Explosive properties</b>                    | Not applicable       |                                |
| <b>Oxidising properties</b>                    | Not applicable       |                                |
| <b><u>Other information</u></b>                |                      |                                |
| <b>Molecular weight</b>                        | Not applicable       |                                |
| <b>VOC content</b>                             | Not applicable       |                                |

**SECTION 10: Stability and reactivity****Reactivity**

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

**Chemical stability**

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****Acute toxicity****Information on likely routes of exposure****Product Information**

.

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

**Numerical measures of toxicity - Product Information**

| Chemical name | Oral LD50                                 | Dermal LD50             | Inhalation LC50                     |
|---------------|-------------------------------------------|-------------------------|-------------------------------------|
| Water         | > 90 mL/kg ( Rat )                        | -                       | -                                   |
| Trade secret  | = 8290 mg/kg ( Rat )                      | > 7940 mg/kg ( Rabbit ) | > 0.83 mg/L ( Rat ) 4 h             |
| Trade secret  | = 22 g/kg ( Rat )                         | > 20 g/kg ( Rabbit )    | -                                   |
| Trade secret  | = 5700 mg/kg ( Rat )<br>= 16 g/kg ( Rat ) | -                       | = 320 mg/m <sup>3</sup> ( Rat ) 4 h |
| Trade secret  | = 27 mg/kg ( Rat )                        | = 20 mg/kg ( Rabbit )   | 0.054 - 0.52 mg/L ( Rat ) 4 h       |

See section 16 for terms and abbreviations



**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

**SECTION 12: Ecological information****Ecotoxicity**

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

**Unknown aquatic toxicity** 0.095 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                 | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| Trade secret  | -                    | LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =5.46mg/L (96h, Pimephales promelas) | -                          | -         |

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** No information available.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | -0.698                |

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

**SECTION 13: Disposal considerations**

**Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information**

**ADG** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

No poisons schedule number allocated

| Chemical name  | National pollutant inventory                                                                                                                                                                                                                            |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trade secret - | 20 MW Threshold category 2b total<br>60000 MWH Threshold category 2b total<br>1 tonne/h Threshold category 2a total<br>25 tonne/yr Threshold category 1a total<br>400 tonne/yr Threshold category 2a total<br>2000 tonne/yr Threshold category 2b total |

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 17-May-2023

**Revision Note**

Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 17-May-2023

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Bio-Plex Standard Diluent  
**Catalogue Number(s)** 9703888, 9704424, 171305042, 171304080M, 10022368, 12005853

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

#### Other hazards which do not result in classification

Contains animal source material (Cattle)

## SECTION 3: Composition/information on ingredients

**Substance**

Not applicable

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Trade secret              | -           | 20 - 35      |
| Trade secret              | -           | 0.3 - 0.99   |
| HBR 1 Blocker             | NO-CAS-18   | 0.1 - 0.299  |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.01 - 0.099 |
| Chemical name             | CAS No      | Weight-%     |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

|                                          |             |
|------------------------------------------|-------------|
| <b>Specific hazards arising from the</b> | None known. |
|------------------------------------------|-------------|

chemical

### Special protective actions for fire-fighters

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## **SECTION 6: Accidental release measures**

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## **SECTION 7: Handling and storage**

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## **SECTION 8: Exposure controls/personal protection**

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

### Individual protection measures, such as personal protective equipment

## SECTION 9: Physical and chemical properties

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks</u> • <u>Method</u> |
|-----------------------------------------|-------------------|--------------------------------|
| pH                                      | 7.4               |                                |
| Melting point / freezing point          | No data available | None known                     |
| Initial boiling point and boiling range | 100 °C            |                                |
| Flash point                             | No data available | None known                     |
| Evaporation rate                        | No data available | None known                     |
| Flammability                            | No data available | None known                     |
| Flammability Limit in Air               |                   | None known                     |
| Upper flammability or explosive limits  | No data available |                                |
| Lower flammability or explosive limits  | No data available |                                |
| Vapour pressure                         | No data available | None known                     |
| Relative vapour density                 | No data available | None known                     |
| Relative density                        | No data available | None known                     |
| Water solubility                        | Miscible in water |                                |
| Solubility(ies)                         | No data available | None known                     |
| Partition coefficient                   | No data available | None known                     |
| Autoignition temperature                | No data available | None known                     |
| Decomposition temperature               |                   | None known                     |
| Kinematic viscosity                     | No data available | None known                     |
| Dynamic viscosity                       | No data available | None known                     |
| Explosive properties                    | Not applicable    |                                |
| Oxidising properties                    | Not applicable    |                                |

|                         |                |
|-------------------------|----------------|
| <b>Molecular weight</b> | Not applicable |
| <b>VOC content</b>      | Not applicable |

## SECTION 10: Stability and reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

**Chemical stability**

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****Acute toxicity****Information on likely routes of exposure****Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

**Numerical measures of toxicity - Product Information**

| Chemical name | Oral LD50                                 | Dermal LD50             | Inhalation LC50                     |
|---------------|-------------------------------------------|-------------------------|-------------------------------------|
| Water         | > 90 mL/kg ( Rat )                        | -                       | -                                   |
| Trade secret  | = 8290 mg/kg ( Rat )                      | > 7940 mg/kg ( Rabbit ) | > 0.83 mg/L ( Rat ) 4 h             |
| Trade secret  | = 22 g/kg ( Rat )                         | > 20 g/kg ( Rabbit )    | -                                   |
| Trade secret  | = 5700 mg/kg ( Rat )<br>= 16 g/kg ( Rat ) | -                       | = 320 mg/m <sup>3</sup> ( Rat ) 4 h |
| Trade secret  | = 27 mg/kg ( Rat )                        | = 20 mg/kg ( Rabbit )   | 0.054 - 0.52 mg/L ( Rat ) 4 h       |

See section 16 for terms and abbreviations



**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

**SECTION 12: Ecological information****Ecotoxicity**

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

**Unknown aquatic toxicity** 0.018 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                 | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| Trade secret  | -                    | LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =5.46mg/L (96h, Pimephales promelas) | -                          | -         |

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** No information available.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | -0.698                |

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

**SECTION 13: Disposal considerations**

**Disposal methods****Waste from residues/unused products**

Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging**

Do not reuse empty containers.

**SECTION 14: Transport information****ADG**

Not regulated

**IATA**

Not regulated

**IMDG**

Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

No poisons schedule number allocated

| Chemical name  | National pollutant inventory                                                                                                                                                                                                                            |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trade secret - | 20 MW Threshold category 2b total<br>60000 MWH Threshold category 2b total<br>1 tonne/h Threshold category 2a total<br>25 tonne/yr Threshold category 1a total<br>400 tonne/yr Threshold category 2a total<br>2000 tonne/yr Threshold category 2b total |

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable**The Stockholm Convention on Persistent Organic Pollutants** Not applicable**The Rotterdam Convention** Not applicable**SECTION 16: Other information****Prepared By**

Bio-Rad Laboratories, Environmental Health and Safety

**Revision date**

17-May-2023

**Revision Note**

Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 17-May-2023

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Streptavidin-PE  
**Catalogue Number(s)** 171304501, 9704418, 9703887, 9703897

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified

### Other hazards which do not result in classification

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Trade secret              | -           | 0.3 - 0.99   |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.01 - 0.099 |
| Avidin                    | 9013-20-1   | 0.01 - 0.099 |
| Chemical name             | CAS No      | Weight-%     |
| Non-hazardous ingredients | Proprietary | Balance      |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                                 |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766             |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>                | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| <b>Skin contact</b>               | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>                  | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

|                                          |             |
|------------------------------------------|-------------|
| <b>Specific hazards arising from the</b> | None known. |
|------------------------------------------|-------------|

chemical

### Special protective actions for fire-fighters

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## **SECTION 6: Accidental release measures**

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## **SECTION 7: Handling and storage**

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## **SECTION 8: Exposure controls/personal protection**

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | Australia                                     | ACGIH TLV                                                                              |
|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| Trade secret  | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

### Individual protection measures, such as personal protective equipment

## SECTION 9: Physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

## Other information

## SECTION 10: Stability and reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

**Chemical stability**

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****Acute toxicity****Information on likely routes of exposure****Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

**Numerical measures of toxicity - Product Information**

| Chemical name | Oral LD50            | Dermal LD50              | Inhalation LC50               |
|---------------|----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )   | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )     | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 8290 mg/kg ( Rat ) | > 7940 mg/kg ( Rabbit )  | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret  | = 27 mg/kg ( Rat )   | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |

See section 16 for terms and abbreviations

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.



|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

## SECTION 12: Ecological information

### Ecotoxicity

#### Ecotoxicity

**Unknown aquatic toxicity** 0.02 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                          | -                                                                                                    |

### Persistence and degradability

**Persistence and degradability** No information available.

### Bioaccumulative potential

**Bioaccumulation** No information available.

### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

## **SECTION 13: Disposal considerations**

**Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## **SECTION 14: Transport information**

**ADG** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

## **SECTION 15: Regulatory information**

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National regulations**

**Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

No poisons schedule number allocated

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

## **SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

Revision date 17-May-2023

Revision Note Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AELG(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**